

ChE 312 Final Exam example questions.

100 points, 3 problems, 5 pages.

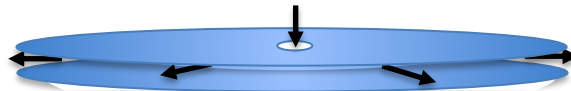
State assumptions; be specific about units; use the right number of significant digits; explain your answers concisely. If you are short of time, be sure at least to explain how to determine the answer. NEVER ERASE! NEVER ERASE! NEVER ERASE! Use ink so you'll NEVER ERASE!

Problem 1 – Short questions (45 points).

(1a, 3 points) In a hot, closed room in the summer, not a breath of air is moving. Suddenly, you smell the faint aroma of perfume. Did it reach you by diffusion or convection?

(1b, 3 points) Downwind of the chicken farm, the smell of ammonia is intense; upwind, you can't smell any ammonia. What is the dominant mode of mass transfer for ammonia here, diffusion or convection?

(1c, 4 points) Gas is injected into the center of a gap between two large, flat disks, and it uniformly flows radially outward.



Modeling the flow as steady-state with reaction and one-dimensional radial convection and diffusion, how would you simplify the following "general continuity equation for species i "?

$$\frac{\partial c_i}{\partial t} + \left(v_r \frac{\partial c_i}{\partial r} + v_\theta \frac{1}{r} \frac{\partial c_i}{\partial \theta} + v_z \frac{\partial c_i}{\partial z} \right) = D_i \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c_i}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 c_i}{\partial \theta^2} + \frac{\partial^2 c_i}{\partial z^2} \right] + R_i$$

(1e, 5 points) What dimensionless number or relationship compares a substance's convective mass-transfer coefficient to its diffusivity, as in mass-transfer-coefficient correlations? Write the relationship.

(1d., 5 points) What dimensionless number or relationship compares the external convective mass-transfer coefficient of a substance to its diffusivity inside an object, as in unsteady-state mass transfer?

(1e., 8 points) An aqueous solution of urea is in laminar flow ($N_{Re}=100$) against a single sphere suspended in the liquid. To calculate the flux of urea from the flowing liquid into an initially urea-free sphere, you will need a convective mass-transfer coefficient. What is a suitable correlation for obtaining it?

(1f., 8 points) Obtaining a convective mass-transfer coefficient for liquid flow in a packed bed of spherical beads is more complicated than the single-sphere case of 1e, so you typically must estimate the mass-transfer coefficient by a correlation. Give an equation showing how the J -factor is related to the mass-transfer coefficient, and give a suitable correlation (supposing $N_{Re}\approx 100$) for obtaining it.

(1g., 9 points) Countercurrent gas-liquid flow in a packed bed of spherical beads is more complicated still, so you typically must obtain the mass-transfer coefficient by a correlation for H_L or H_G . Give an equation showing how H_L is related to the mass-transfer coefficient, and give a suitable correlation for obtaining it.

Problem 2 (10 points).

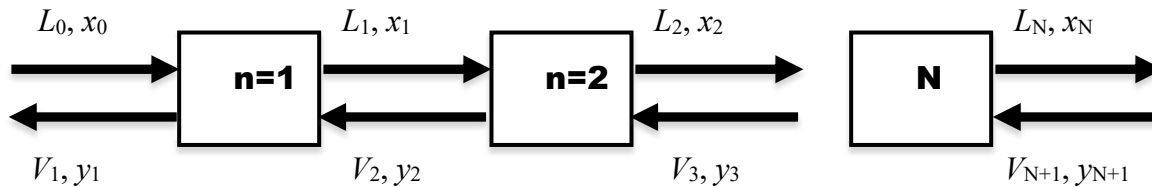
(2a, 5 pts) What is the difference between mass-transfer coefficients k and k' ?

(2b, 5 pts) For mass transfer across an interface, convective mass-transfer resistance may be significant on either or both sides. For example, at a gas-liquid interface, $y_{A,\text{interface}}$ may not be the same as $y_{A,\text{bulk}}$, and $x_{A,\text{interface}}$ may not be the same as $x_{A,\text{bulk}}$.

How are $y_{A,\text{interface}}$ and $x_{A,\text{interface}}$ related to each other at steady state?

Problem 3 (45 points).

Consider countercurrent liquid-liquid separation using equilibrium contacting stages. A feed liquid containing the solute (input rate L_0 and solute mole fraction x_0) contacts another immiscible liquid (flow and solute mole fraction labeled V_2, y_2) in the first equilibrium stage, $n=1$. The equilibrium relation for solute is $(\gamma_y y)_n = (\gamma_x x)_n$ (assume you have equations for finding γ_y and γ_x).



We can analyze this staged equilibrium separation with the same general approach that was used for aequilibrium-stage absorption, stripping, and distillation.

(3a, 5 points each) Write the steady-state solute material balance around stage 1.

(3b, 5 points) Using $n=N$ for the last stage, fed countercurrently by V_{N+1}, y_{N+1} , write the steady-state solute material balance for the overall system (stages 1 through N).

(3b, 5 points) Write the material balance around the set of stages 1 through n and express it as an "operating-line equation"; that is, as $y_{n+1} = f(x_n)$.

(3c, 20 pts) Describe the steps you would use to calculate the number of stages to achieve mole fraction y_I if you are given a solute mole fraction specification y_I ? What assumptions are necessary for L and V ?

(3d, 10 pts) Describe the stagewise analysis steps used in the McCabe-Thiele algorithm (binary mixture, total condenser, partial reboiler, feed quality q), assuming known feed and distillate compositions. Compare and contrast them to the steps used in 3c.